

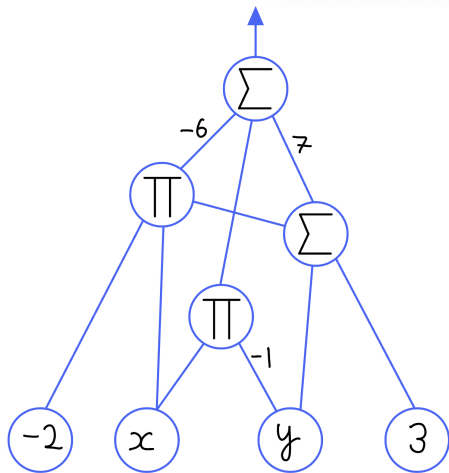
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Debordering Closure Results in Determinantal and Pfaffian Ideals

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Algebraic Circuits



We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input-output path

Figure: $-6(-2x(y+3)) - xy + 7(x+3)$

Algebraic Circuits with Oracles

Instead of just allowing Σ and Π gates, we can take any polynomial $f(\bar{x})$ and use this as a *f -oracle* gate.

For example, if $f(\bar{x}) = r(\bar{x}) \cdot g(\bar{x})$, we can ask about the circuit complexity for g if we allow an f -oracle gate.

Circuit Complexity for Ideals

In algebraic complexity, we are interested in describing the *circuit complexity* of some family of polynomials.

Definition

Let $\mathbb{F}[\bar{x}]$ be the set of polynomials with variables $\bar{x} = x_1, \dots, x_n$ and coefficients in $\mathbb{F}$.

A subset $I \subseteq \mathbb{F}[\bar{x}]$ is called an *ideal* if it satisfies

1. $f(\bar{x}), g(\bar{x}) \in I \implies f(\bar{x}) + g(\bar{x}) \in I$
2. $f(\bar{x}) \in I, r(\bar{x}) \in \mathbb{F}[\bar{x}] \implies r(\bar{x}) \cdot f(\bar{x}) \in I$

Circuit Complexity for Ideals

Theorem (Hilbert's Basis Theorem)

If $I \subseteq \mathbb{F}[\bar{x}]$ is an ideal, then there exist finitely many polynomials $g_1(\bar{x}), \dots, g_k(\bar{x}) \in I$ which *generate* I .

$$f(\bar{x}) \in I \implies f(\bar{x}) = \sum_{i=1}^k h_i(\bar{x}) \cdot g_i(\bar{x}), \quad h_i(\bar{x}) \in \mathbb{F}[\bar{x}].$$

We will write $I = \langle g_1, \dots, g_k \rangle$.

Question

Suppose $f \in \langle g_1, \dots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$?

Principal Ideals

Example

The *principal ideals* are generated by a single polynomial, so $I = \langle g \rangle$. If $f \in \langle g \rangle$, then g is a *factor* of f .

Question

Suppose $I = \langle g \rangle$. Let $f \in I$. How does the complexity of f compare to the complexity of g ?

In other words, if g is a factor of f and f has a small circuit, does g have a small circuit?

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If g is a factor of f , $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$.

Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0, g can be *approximately computed* by a circuit of size $\text{poly}(\text{size}(f), \deg(f))$.

By approximate computation, we mean the circuit computes a polynomial $h(\bar{x}) \in \mathbb{F}(\varepsilon)[\bar{x}]$ such that $h(\bar{x}) - g(\bar{x}) \in \varepsilon \mathbb{F}[[\varepsilon]][\bar{x}]$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Determinantal Ideals

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Conjecture ([Gro, Conjecture 6.3])

For every nonzero polynomial f in $I_{n,n,n/2}$, there is a constant-depth f -oracle algebraic circuit of size $\text{poly}(n)$ which computes the $t \times t$ determinant for some $t = n^{\Theta(1)}$.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *approximately computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

- We also prove analogous results for *Pfaffian* ideals in the case that our matrix of variables X is skew-symmetric ($x_{j,i} = -x_{i,j}$).
- We need that our fields have size $\text{poly}(\deg(f), n, m)$ as well as characteristic greater than $\deg(f)$.

Bideterminants

We will first discuss which generators we will use for $I_{n,m,r}^{\det}$.

Definition (Young Tableau)

A *partition* is a sequence of integers $\sigma = (\sigma_1 \geq \cdots \geq \sigma_k \geq 0)$.

A *Young Tableau* of shape σ is a filling of the squares in the diagram for σ using positive integers.

Such a filling is *conjugate semistandard* if it is strictly increasing down columns and weakly increasing along rows.

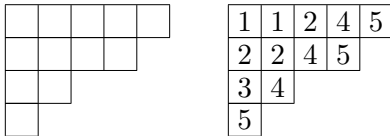


Figure: Diagram for $\sigma = (5, 4, 2, 1)$ and a Young Tableau of shape σ

Bideterminants

We have *canonical* semistandard tableau.

1	2	3	4	5
1	2	3	4	
1	2			
1				

Figure: Canonical tableau of shape $\sigma = (5, 4, 2, 1)$

We will notate the canonical semistandard tableau of shape σ by K_σ .

Bideterminants

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$ and $(S \mid T)$ is the *bitableau*

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 & & \\ \hline 4 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 & & \\ \hline 3 & & & \\ \hline \end{array} \right),$$

then the *bideterminant* $(S \mid T)(X)$ is given by

$$(S \mid T)(X) = \begin{pmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{pmatrix} \begin{pmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{pmatrix} (x_{4,3}).$$

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A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ . Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B} (A \mid B)(X),$$

where A and B are *conjugate semistandard*.

Homogenization

Definition

Consider a degree d polynomial $f \in \mathbb{F}[\bar{x}, t]$. Then we have polynomials $\text{coeff}_{t^0}(f), \dots, \text{coeff}_{t^d}(f) \in \mathbb{F}[\bar{x}]$:

$$f(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(f) t^i$$

Lemma (Folklore)

Suppose that f is computed by a g -oracle circuit of size s and depth Δ . Then for each $0 \leq i \leq d$, we can compute $\text{coeff}_{t^i}(f)$ using a g -oracle circuit of size $O(ds)$, depth $\Delta + 1$.

This circuit is computing via interpolation at $d + 1$ *distinct* points in $\mathbb{F}$. Furthermore, if the top gate of the circuit for f is a Σ -gate, then $\text{coeff}_{t^i}(f)$ can be computed with depth Δ .

Isolation Lemma

Extracting homogeneous components will not be enough, as different bideterminants may have the same total degree. However, a *random* monomial substitution will allow us to isolate a single term.

Lemma ([KS01, Lemma 4])

Consider a set $\mathcal{L}$ of linear forms in variables $e_1, \dots, e_\ell$ with integer coefficients between 0 and K .

Fix $\varepsilon > 0$. Choose $z_1, \dots, z_\ell$ independently and uniformly at random from $\{0, \dots, M\}$, where $M \geq K\ell/\varepsilon$, and evaluate all forms in $\mathcal{L}$ at $z_1, \dots, z_\ell$.

Then with probability at least $1 - \varepsilon$, there is a *unique* form in $\mathcal{L}$ attaining the minimum value.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $f(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in f is at most K :

$$f(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Fix $\varepsilon > 0$. Choose $z_1, \dots, z_\ell$ independently and uniformly at random from $\{0, \dots, M\}$, where $M \geq K\ell/\varepsilon$, and define a morphism

$$x_i \mapsto w^{z_i}.$$

Then with probability at least $1 - \varepsilon$, there is a *unique* monomial of f with which attains the minimum degree in w under this morphism.

Issues with Homogenization

If a circuit approximately computes $f(\bar{x})$, then it computes

$$f(\bar{x}) + \varepsilon^q g(\bar{x}, \varepsilon)$$

for some $q \geq 0$.

q can be *arbitrarily large* $\implies$ homogenization gives *large circuit*.

An equivalent phrasing of our Isolation Lemma is that there *exists* an assignment $x_i \mapsto w^{z_i}$ with a unique monomial of lowest degree, where the total degree in w is not too large.

Using the Isolation Lemma and Homogenization, we extract a *single* monomial and compute this monomial with a small circuit.

Proof Outline

Since we are starting with $f \in I_{n,m,r}^{\det}$, we can express in terms of our standard basis:

$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X)$$

Our goal will be to use $f(X)$ as an oracle to compute a canonical bideterminant.

Proof Outline

$$\sum_{k \in I} c_k(S_k \mid T_k)(X)$$

We apply a linear transformation to X to get an expression of the form

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k}(K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} b_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell}(S_\ell \mid T_\ell)(X)$$

- The Λ and Ξ are extra variables “tracking” the changes made by our linear transformation which send the $(S_k \mid T_k) \mapsto (K_{\sigma_k} \mid K_{\sigma_k})$.
- The terms indexed by $\ell \in B$ are “garbage terms” produced by this linear map.

Proof Outline

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} b_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} (S_\ell \mid T_\ell)(X)$$

We do another linear transformation to get an expression of the form

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} v^{D_k} (K_{\sigma_k} \mid K_{\sigma_k})(X) + \text{terms of lower degree in } v$$

Homogenization $\implies$ f -oracle circuit for

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Proof Outline

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

All of these terms now have the “same total degree” so we cannot homogenize to keep a single one.

Apply a random map

$$\lambda_{a,b} \mapsto w^{t_{a,b}}$$

$$\xi_{i,j} \mapsto w^{t_{i,j}}$$

$$\sum_{k \in A} a_k w^{\bar{t}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Proof Outline

$$\sum_{k \in A} a_k w^{\bar{t}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Isolation Lemma $\implies$ there is a choice of the $t_{i,j}$ not too large so that every term has *distinct* degree in w .

We can then use Homogenization again to extract a single term.

$$a(K_{\sigma} \mid K_{\sigma})(X)$$

Thank You!

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